

Discovering/Capturing a Mobile on a Line (Draft Notes)

Evangelos Kranakis¹³

¹ School of Computer Science, Carleton University, Ottawa, Ontario, Canada.

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Abstract. A mobile is moving away from the origin with speed $v < 1$ starting at an unknown distance d from the origin. A robot starts at the same time from the origin and wants to either discover or capture the mobile. We give an algorithm for capturing the mobile and analyze its competitive ratio.

Key words and phrases.

1 Introduction

1.1 Similarities between Search on a Line and Spiral-Search

Recall that in standard *linear search* on an infinite line the searcher stays always on the line throughout the search for the target, be that either static or mobile. One can imagine a search model on an infinite line that has similarities to spiral search in the plane. Indeed, assume there is a target (either static or oblivious mobile) on the infinite line and that the searcher is riding an airplane. The target is *discovered* only if the searcher has landed on the line and can see that the target is located between the origin and the position of the searcher. The target is *captured* only if it is discovered by the searcher and the searcher can meet the target by moving towards it.

In a way, we think that the searcher is “forward blind” in that it cannot see the target if the latter lies ahead of the searcher! It can see it only if it is located between the searcher and the base station at the origin. In the sequel we explore these discovery and capture models on an infinite line.

2 Search on an Infinite Line

We consider an infinite line with a specified origin. Table 1 displays a summary of what we can prove.

Table 1. Upper and lower bounds for linear-, discovery-, and capture-search, as well search for an oblivious mobile target moving away from the origin with constant speed $v < 1$, where $a^* = \frac{3+4v-v^3+\sqrt{v^6+8v^4+10v^3+8v+9}}{4(1-v)}$, is the corresponding growth ratio of capture search.

	CR_{lis}	CR_{dis}	CR_{cap}
UB	9	$\frac{11+5\sqrt{5}}{2}$	$\frac{25}{2}$
LB	9	$\frac{11+5\sqrt{5}}{2}$	$\frac{25}{2}$
	$CR_{lis}(v)$	$CR_{dis}(v)$	$CR_{cap}(v)$
UB	$1 + \frac{8(1+v)}{(1-v)^2}$	$\frac{(\sqrt{5}+8v-1)(\sqrt{5}+8v+3)^3}{16(1-v)}$	$\frac{1}{1+v} \left(-1 + \frac{(a^*)^2(a^*-v-v^2)}{(1-v)a^*-(1+v)} \right)$
LB	$1 + \frac{8(1+v)}{(1-v)^2}$	??	??

2.1 Static Target: Upper Bounds

Consider the standard zig-zag algorithm with growth ratio $a > 1$ and suppose the target is at distance d from the origin. Consider the first k such that $a^k < d \leq a^{k+1}$. In standard linear search, at the end of the $(k-1)$ -1st round is at the origin at time

$$2(a^0 + a^1 + \dots + a^{k-1}) = 2\frac{a^k - 1}{a - 1}.$$

Now it moves distance a^k but misses the target since $a^k < d$. It returns to the origin goes in the opposite direction for distance a^{k+1} and returns to the origin in additional time a^{k+1} . At the time the robot returns to the the total time is

$$2(a^0 + a^1 + \dots + a^{k+1}) = 2\frac{a^{k+2} - 1}{a - 1}.$$

2.1.1 Linear Search. Since we are analyzing linear search, it will catch the target in additional time d . The competitive ratio will be

$$\begin{aligned}
CR_{lis} &< \frac{2\frac{a^{k+2}-1}{a-1} + d}{d} \\
&< 1 + \frac{2\frac{a^{k+2}-1}{a-1}}{a^k} \\
&< 1 + \frac{2\frac{a^{k+2}}{a-1}}{a^k} \\
&= 1 + 2\frac{a^2}{a-1}.
\end{aligned}$$

The minimizer and the corresponding minimum of the upper bound above thus obtained are

$$\boxed{a = 2 \text{ and } 9}$$

2.1.2 Discover Search. The discovery model has the same analysis except that in the last move it lands at position a^{k+2} . It follows that

$$\begin{aligned}
CR_{dis} &< \frac{2\frac{a^{k+2}-1}{a-1} + a^{k+2}}{d} \\
&< \frac{2\frac{a^{k+2}-1}{a-1} + a^{k+2}}{a^k} \\
&< \frac{2\frac{a^{k+2}}{a-1} + a^{k+2}}{a^k} \\
&< a^2 + 2\frac{a^2}{a-1}.
\end{aligned}$$

Note that

$$f(a) = a^2 + 2\frac{a^2}{a-1} = a^2 + 2a + 2 + \frac{2}{a-1}.$$

Its derivative is $f'(a) = 2a + 2 - \frac{2}{(a-1)^2}$. Set $f'(a) = 0$ to get the equation $a^3 - a^2 - a = 0$, which reduces to $a^2 - a - 1 = 0$. The minimizer and the corresponding minimum of the upper bound above thus obtained are

$$\boxed{\frac{1+\sqrt{5}}{2} \text{ and } \frac{11+5\sqrt{5}}{2} \approx 11.0901 \dots}$$

2.1.3 Capture Search. The capture model has the same analysis except that in the last move it will see the target only when it lands at position a^{k+3} . After seeing the target it will move back distance $a^{k+2} - d$ to capture the target. It follows that

$$\begin{aligned}
CR_{cap} &< \frac{2\frac{a^{k+2}-1}{a-1} + a^{k+2} + a^{k+2} - d}{d} \\
&< \frac{2\frac{a^{k+2}-1}{a-1} + 2a^{k+2}}{d} - 1 \\
&< \frac{2\frac{a^{k+2}}{a-1} + 2a^{k+2}}{a^k} - 1 \\
&= 2\left(\frac{a^2}{a-1} + a^2\right) - 1 \\
&= 2a^2\left(1 + \frac{1}{a-1}\right) - 1.
\end{aligned}$$

The derivative of the function

$$f(a) = 2a^2\left(1 + \frac{1}{a-1}\right),$$

is equal to

$$f'(a) = 2\left(2a + 1 - \frac{1}{(a-1)^2}\right).$$

The minimizer and the corresponding minimum of the upper bound above thus obtained are

$$a = \frac{3}{2} \text{ and } \frac{25}{2}$$

2.2 Static Target: Lower Bounds

Let the searcher start at 0 and move at unit speed. Let the target be at unknown position $x \in \mathbb{R}$. A deterministic strategy is C -competitive if it finds every target at distance $|x|$ in time at most $C|x|$.

2.2.1 Linear Search. We prove $C \geq 9$.

Consider any deterministic C -competitive strategy. Ignore wasted motion and list its successive turning points as $0 \rightarrow +x_1 \rightarrow -x_2 \rightarrow +x_3 \rightarrow -x_4 \rightarrow \dots$, where $x_i > 0$ and $\lim_{i \rightarrow +\infty} x_i = +\infty$. Put

$$S_i = \sum_{j=1}^i x_j \text{ and } s_i = \frac{S_{i-1}}{x_i}. \quad (1)$$

Let T_i be the time at which the searcher reaches the i -th turning point: $T_i = 2S_{i-1} + x_i$.

Place the target at a point x just beyond the i -th turning point, at distance $x = x_i + \epsilon$ on the same side, for some $\epsilon > 0$ arbitrarily small. It is missed at time T_i . The next time the searcher reaches that side, it must go to the opposite turning point $-x_{i+1}$ and back. **Upon its return to the origin it will travel a distance $x_i + \epsilon$ until it discovers the target.** Hence, the finding time is $T_i + x_i + 2x_{i+1} + x_i + \epsilon = 2S_{i-1} + 3x_i + 2x_{i+1} + \epsilon$. Since the strategy is C -competitive,

$$C \geq \frac{2S_{i-1} + 3x_i + 2x_{i+1}}{x_i} = 3 + 2\frac{x_{i+1}}{x_i} + 2s_i.$$

Thus

$$\frac{x_{i+1}}{x_i} \leq \alpha - s_i, \text{ where } \alpha = \frac{C-3}{2}. \quad (2)$$

Therefore

$$s_{i+1} = \frac{S_i}{x_{i+1}} = \frac{S_{i-1} + x_i}{x_{i+1}} = \frac{S_{i-1}/x_i + 1}{x_{i+1}/x_i} = \frac{s_i + 1}{x_{i+1}/x_i} \geq \frac{s_i + 1}{\alpha - s_i}. \quad (3)$$

For the sake of contradiction, assume that $C < 9$. Then $\alpha < 3$. Considering the righthand side of Inequality (3), define the function

$$f(s) := \frac{s+1}{\alpha-s}, \text{ for } 0 \leq s < \alpha.$$

Observe that for $\alpha < 3$ the quadratic $s^2 + (1-\alpha)s + 1$ is positive. Hence,

$$f(s) - s = \frac{s^2 + (1-\alpha)s + 1}{\alpha-s} > 0. \quad (4)$$

Inequalities (3) and (4) yield $s_{i+1} \geq f(s_i) > s_i$. So the sequence $(s_i : i \geq 0)$ defined in Equation (1) is strictly increasing. Also, we observe from Inequality (2) that $s_i < \alpha$ for all i , since $0 < \frac{x_{i+1}}{x_i} \leq \alpha - s_i$.

Thus the sequence $(s_i : i = 1, 2, \dots)$ is increasing and bounded above by α , so it converges to some $L \leq \alpha$. Since $s_i < \alpha$ for all i , actually $L < \alpha$. Passing to the limit in $s_{i+1} \geq f(s_i)$ gives $L \geq f(L) > L$, a contradiction. Therefore $C < 9$ is impossible, so every deterministic strategy satisfies $C \geq 9$.

2.2.2 Discovery Search. Consider any deterministic C -competitive strategy. Ignore wasted motion and list its successive turning points as $0 \rightarrow +x_1 \rightarrow -x_2 \rightarrow +x_3 \rightarrow -x_4 \rightarrow \dots$, where $x_i > 0$. Without loss of generality we may assume $x_i < x_{i+2}$, for all i . Put

$$S_i = \sum_{j=1}^i x_j \text{ and } s_i = \frac{S_{i-1}}{x_i}. \quad (5)$$

The time T_i at which the searcher reaches the i -th turning point is $T_i = 2S_{i-1} + x_i$.

Place the target just beyond the i -th turning point, at distance $x_i + \varepsilon$ on the same side. It is missed at time T_i . The next time the searcher reaches that side, it must go to the opposite turning point $-x_{i+1}$ and back. **Upon its return to the origin it will travel a distance x_{i+2} until it discovers the target.** Hence, the discovering time is $T_i + x_i + 2x_{i+1} + x_{i+2} = 2S_{i-1} + 2(x_i + x_{i+1}) + x_{i+2}$. Since $x_i < x_{i+2}$, for all i , and the strategy is C -competitive,

$$\begin{aligned} C &\geq \frac{2S_{i-1} + 2(x_i + x_{i+1}) + x_{i+2}}{x_i} \\ &= 2s_i + 2 + \frac{2x_{i+1} + x_{i+2}}{x_i} \\ &> 2s_i + 2 + 2\frac{x_{i+1}}{x_i} + \frac{x_{i+2}}{x_{i+1}} \cdot \frac{x_{i+1}}{x_i} \\ &> 2s_i + 2 + \frac{x_{i+1}}{x_i} \left(2 + \frac{x_{i+2}}{x_{i+1}} \right). \end{aligned}$$

Thus,

$$\frac{x_{i+1}}{x_i} \left(2 + \frac{x_{i+2}}{x_{i+1}} \right) \leq \alpha - s_i, \text{ where } \alpha = \frac{C-2}{2}.$$

It follows that assuming $x_{i+2}/x_{i+1} \geq 1$ we have

$$3\frac{x_{i+1}}{x_i} \leq \alpha - s_i, \text{ where } \alpha = \frac{C-2}{2}. \quad (6)$$

Therefore,

$$s_{i+1} = \frac{S_i}{x_{i+1}} = \frac{S_{i-1} + x_i}{x_{i+1}} = \frac{S_{i-1}/x_i + 1}{x_{i+1}/x_i} = \frac{s_i + 1}{x_{i+1}/x_i} \geq \frac{3(s_i + 1)}{\alpha - s_i}. \quad (7)$$

2.2.3 Capture Search. Consider any deterministic C -competitive strategy. Ignore wasted motion and list its successive turning points as $0 \rightarrow +x_1 \rightarrow -x_2 \rightarrow +x_3 \rightarrow -x_4 \rightarrow \dots$, where $x_i > 0$. Without loss of generality we may assume $x_i < x_{i+2}$, for all i . Put

$$S_i = \sum_{j=1}^i x_j \text{ and } s_i = \frac{S_{i-1}}{x_i}. \quad (8)$$

The time T_i at which the searcher reaches the i -th turning point is $T_i = 2S_{i-1} + x_i$.

Place the target at a point x just beyond the i -th turning point, at distance $x = x_i + \epsilon$ on the same side, for some $\epsilon > 0$ arbitrarily small. It is missed at time T_i . The next time the searcher reaches that side, it must go to the opposite turning point $-x_{i+1}$ and back. **Upon its return to the origin it will travel a distance x_{i+2} until it discovers the target and an additional $x_{i+2} - x_i + \epsilon$ to capture it.** Hence, the finding time is $T_i + x_i + 2x_{i+1} + x_{i+2} + x_{i+2} - x_i + \epsilon = 2S_{i-1} + x_i + 2x_{i+1} + 2x_{i+2} - \epsilon$. Since the strategy is C -competitive,

$$\begin{aligned} C &\geq \frac{2S_{i-1} + x_i + 2x_{i+1} + 2x_{i+2} - \epsilon}{x_i} \\ &= 2s_i + 1 + 2\frac{x_{i+1} + x_{i+2} - \epsilon}{x_i} \\ &= 2s_i + 1 + 2\frac{x_{i+1}}{x_i} + 2\frac{x_{i+2}}{x_{i+1}} \cdot \frac{x_{i+1}}{x_i} - \frac{2\epsilon}{x_i} \\ &= 2s_i + 1 + 2\frac{x_{i+1}}{x_i} \left(1 + \frac{x_{i+2}}{x_{i+1}}\right) - \frac{2\epsilon}{x_i} \\ &\geq 2s_i + 1 + 4\frac{x_{i+1}}{x_i} - \frac{2\epsilon}{x_i} \quad \boxed{\text{(assuming } x_{i+2} \geq x_{i+1})} \end{aligned}$$

Thus,

$$\frac{x_{i+1}}{x_i} \leq \alpha - s_i/2 + \frac{\epsilon}{2x_i}, \text{ where } \alpha = \frac{C-1}{4}. \quad (9)$$

Therefore

$$s_{i+1} = \frac{S_i}{x_{i+1}} = \frac{S_{i-1} + x_i}{x_{i+1}} = \frac{S_{i-1}/x_i + 1}{x_{i+1}/x_i} = \frac{s_i + 1}{x_{i+1}/x_i} > \frac{s_i + 1}{\alpha - s_i/2}, \quad (10)$$

where $\alpha = \frac{C-1}{4}$.

2.3 Oblivious Mobile Target: Upper Bounds

The argument generalizes to the case of an oblivious mobile target moving away from the origin with constant speed $0 \leq v < 1$ away from the origin. Consider the zig-zag algorithm with growth ratio a . We know that

$$a = 2\frac{1+v}{1-v}$$

is the optimizer for linear search with such an oblivious mobile target. We would like to redo the proof and use it to compute the competitive ratios for $CR_{dis}(v)$ and $CR_{cap}(v)$.

Zig-zag search is in rounds $0, 1, \dots, k, \dots$. Assume the robot misses the target in round k , but captures it in round $k+2$. Let T_k be the time at the origin upon completion of round $k-1$. Clearly,

$$T_k = 2a^0 + 2a^1 + 2a^{k-1} = 2\frac{a^k - 1}{a - 1}.$$

The robot is at the origin at time T_k while the mobile is at distance $d + vT_k$; if the robot misses the mobile when it moves additional distance a^k the reason is that

$$a^k < \frac{d + vT_k}{1 - v}. \quad (11)$$

Let k be maximal such that Inequality (11) is valid. The robot now returns to the origin in additional time a^k , moves in the opposite direction distance a^{k+1} and returns back to origin. The total time when the robot returns again to the origin will be now

$$T_{k+2} = T_k + 2a^k + 2a^{k+1} = 2\frac{a^{k+2} - 1}{a - 1}.$$

The maximality of k in Inequality (11) implies that

$$a^{k+2} \geq \frac{d + vT_{k+2}}{1 - v}. \quad (12)$$

2.3.1 Linear Search. In linear search the robot will catch the target in additional time $\frac{d + vT_{k+2}}{1 - v}$, where $a^{k+2} \geq \frac{d + vT_{k+2}}{1 - v}$. Therefore, using Inequality (11),

$$\begin{aligned} CR_{lis}(v) &= \frac{T_{k+2} + \frac{d + vT_{k+2}}{1 - v}}{\frac{d}{1 - v}} = 1 + \frac{T_{k+2}}{d} < 1 + \frac{T_{k+2}}{a^k(1 - v) - vT_k} \\ &= 1 + \frac{2\frac{a^{k+2} - 1}{a - 1}}{a^k(1 - v) - 2v\frac{a^k - 1}{a - 1}} = 1 + \frac{2(a^{k+2} - 1)}{a^k(a - 1)(1 - v) - 2v(a^k - 1)} \\ &< 1 + \frac{2a^{k+2}}{a^k(a - 1)(1 - v) - 2v(a^k - 1)} < 1 + \frac{2a^{k+2}}{a^k(a - 1)(1 - v) - 2va^k} \\ &= 1 + \frac{2a^2}{(a - 1)(1 - v) - 2v}. \end{aligned}$$

Note that the condition $a > \frac{1+v}{1-v}$ (which is equivalent to $(a - 1)(1 - v) - 2v > 0$) is required to ensure that the denominator above is > 0 . The minimizer and the corresponding minimum of the upper bound above thus obtained are

$$a = 2\frac{1+v}{1-v} \text{ and } 1 + \frac{8(1+v)}{(1-v)^2}$$

For linear search, we do know that $1 + \frac{8(1+v)}{(1-v)^2}$ is also a lower bound.

2.3.2 Discover Search. In the discover model, the robot will move (fly) additional distance a^{k+2} , which will be ahead of the mobile since $a^{k+2} \geq \frac{d+vT_{k+2}}{1-v}$. Therefore,

$$\begin{aligned}
CR_{dis}(v) &= \frac{T_{k+2} + \frac{a^{k+2}}{1-v}}{\frac{d}{1-v}} = \frac{2\frac{a^{k+2}-1}{a-1} + \frac{a^{k+2}}{1-v}}{\frac{d}{1-v}} \\
&< \frac{2\frac{a^{k+2}}{a-1} + \frac{a^{k+2}}{1-v}}{\frac{a^k(1-v)-v2\frac{a^k-1}{a-1}}{1-v}} \\
&< \frac{2a^{k+2}(1-v) + a^{k+2}(a-1)}{a^k(1-v)(a-1) - 2v(1-v)a^k} \quad (\text{multiply by } (a-1)(1-v)) \\
&< \frac{2a^2(1-v) + a^2(a-1)}{(1-v)(a-1) - 2v(1-v)} \\
&= \frac{a^3 + a^2(1-2v)}{(1-v)(a-1) - 2v(1-v)} \\
&= \frac{a^3 + a^2(1-2v)}{(1-v)((a-1) - 2v)} \\
&= \frac{a^2(a+1-2v)}{(1-v)(a-1-2v)}.
\end{aligned}$$

Clearly, the term $a+1-2v$ must be > 0 since $a > 1$. The denominator $a-1-2v$ is bigger than 0 iff $a > 1+2v$. The minimizer, under the assumption $a > 1+2v$, and the corresponding minimum of the upper bound above thus obtained are

$$a = \frac{1+4v+\sqrt{5+8v}}{2} \quad \text{and} \quad \frac{(\sqrt{5+8v}-1)(\sqrt{5+8v}+3)^3}{16(1-v)}$$

Note that the last two formulas above match the corresponding formulas for the static case $v = 0$.

2.3.3 Capture Search. In the capture model, just like the discover model, the robot will move (fly) additional distance a^{k+2} , which will be ahead of the mobile, since

$$a^{k+2} \geq \frac{d+vT_{k+2}}{1-v}.$$

In addition, however, the robot will turn back and travel additional distance

$$\frac{a^{k+2} - \frac{d+vT_{k+2}}{1-v}}{1+v}$$

in order to capture the mobile. Therefore, using Inequality (11),

$$\begin{aligned}
CR_{cap}(v) &= \frac{T_{k+2} + \frac{a^{k+2}}{1-v} + \frac{a^{k+2} - \frac{d+vT_{k+2}}{1-v}}{1+v}}{\frac{d}{1-v}} \\
&= \frac{T_{k+2} + \frac{2a^{k+2}}{1-v^2} - \frac{d+vT_{k+2}}{1-v^2}}{\frac{d}{1-v}} \\
&= -\frac{1}{1+v} + \frac{T_{k+2} + \frac{2a^{k+2}}{1-v^2} - \frac{vT_{k+2}}{1-v^2}}{\frac{d}{1-v}} \\
&= -\frac{1}{1+v} + \frac{\frac{2a^{k+2}}{1-v^2} + \frac{(1-v-v^2)T_{k+2}}{1-v^2}}{\frac{d}{1-v}} \\
&= -\frac{1}{1+v} + \frac{2a^{k+2} + (1-v-v^2)T_{k+2}}{d(1+v)} \\
&= \frac{1}{1+v} \left(-1 + \frac{2a^{k+2} + (1-v-v^2)T_{k+2}}{d} \right).
\end{aligned}$$

Next we substitute T_k, T_{k+2}

$$\begin{aligned}
CR_{cap}(v) &< \frac{1}{1+v} \left(-1 + \frac{2a^{k+2} + 2(1-v-v^2)\frac{a^{k+2}}{a-1}}{d} \right) \\
&< \frac{1}{1+v} \left(-1 + \frac{2a^{k+2} + 2(1-v-v^2)\frac{a^{k+2}}{a-1}}{a^k(1-v) - vT_k} \right) \\
&< \frac{1}{1+v} \left(-1 + \frac{2a^{k+2} + 2(1-v-v^2)\frac{a^{k+2}}{a-1}}{a^k(1-v) - 2v\frac{a^k}{a-1}} \right) \\
&< \frac{1}{1+v} \left(-1 + \frac{2a^2 + 2(1-v-v^2)\frac{a^2}{a-1}}{(1-v) - \frac{2v}{a-1}} \right) \\
&< \frac{1}{1+v} \left(-1 + 2\frac{a^2(a-1) + (1-v-v^2)a^2}{(1-v)(a-1) - 2v} \right) \\
&= \frac{1}{1+v} \left(-1 + 2\frac{a^2(a-v-v^2)}{(1-v)(a-1) - 2v} \right) \\
&= \frac{1}{1+v} \left(-1 + 2\frac{a^2(a-v-v^2)}{(1-v)a - (1+v)} \right).
\end{aligned}$$

For $0 \leq v < 1$ and $a > \frac{1+v}{1-v}$ (note that the condition $a > \frac{1+v}{1-v}$ is required to ensure the denominator of the last term above is > 0). The minimizer and the corresponding minimum of the upper bound above thus obtained are

$$a^* = \frac{3+4v-v^3+\sqrt{v^6+8v^4+10v^3+8v+9}}{4(1-v)} \text{ and } \frac{1}{1+v} \left(-1 + 2\frac{(a^*)^2(a^*-v-v^2)}{(1-v)a^*-(1+v)} \right)$$

Note that $a^* = \frac{3}{2}$, for $v = 0$, which is consistent with the formula for the static case.³

2.4 Oblivious Mobile Target: Lower Bounds

Can we prove lower bounds for the proceeding models of liner, discover, and a capture search on an infinite line?

2.4.1 Linear Search. Let the searcher start at 0 and move at unit speed. The target starts at an unknown position $x > 0$ and moves to the right with constant speed $v < 1$. If its initial position were known, the optimal capture time would be $\frac{x}{1-v}$. A deterministic strategy is C -competitive if it captures every such target in time at most $C\frac{x}{1-v}$. Consider any deterministic strategy. List its successive turning points as

$$0 \rightarrow +x_1 \rightarrow -x_2 \rightarrow +x_3 \rightarrow -x_4 \rightarrow \cdots, \quad x_i > 0.$$

Define

$$S_i = \sum_{j=1}^i x_j, \quad s_i = \frac{S_{i-1}}{x_i}, \quad r_i = \frac{x_{i+1}}{x_i}.$$

The time at which the searcher reaches the i -th turning point on the positive side is $T_i = 2S_{i-1} + x_i$.

Assume the target is on the positive side and moves to the right. At time T_i , the searcher is at $+x_i$. For the target to be just missed at that moment, its initial position must be

$$x = x_i - vT_i + \varepsilon$$

with $\varepsilon > 0$. If $x_i - vT_i \leq 0$, then any sufficiently small $x > 0$ is missed at T_i , and the competitive ratio becomes arbitrarily large; hence a finite competitive ratio requires $x_i - vT_i > 0$ for all relevant i .

After time T_i , the searcher goes to $-x_{i+1}$ and then turns right. The capture time T_c satisfies

$$T_c = T_i + \frac{x + vT_i + x_i + 2x_{i+1}}{1 - v}.$$

Since the strategy is C -competitive,

$$T_c \leq C \frac{x}{1 - v}.$$

Multiplying by $1 - v$, substituting $x = x_i - vT_i + \varepsilon$, and letting $\varepsilon \rightarrow 0$, gives

$$T_i + x_i + 2x_{i+1} \leq (C - 1)(x_i - vT_i).$$

Using $T_i = 2S_{i-1} + x_i$, we obtain

$$s_i + 1 + r_i \leq (C - 1) \left(\frac{1 - v}{2} - vs_i \right).$$

³ **N.B.** Formula was found by an AI but I did not even try to verify it.

Thus

$$r_i \leq A - Bs_i,$$

where

$$A = \frac{(C-1)(1-v)}{2} - 1, \quad B = 1 + (C-1)v.$$

Therefore

$$s_{i+1} = \frac{S_i}{x_{i+1}} = \frac{S_{i-1} + x_i}{x_{i+1}} = \frac{s_i + 1}{r_i} \geq \frac{s_i + 1}{A - Bs_i}.$$

Define

$$f(s) = \frac{s+1}{A-Bs}.$$

Then

$$s_{i+1} \geq f(s_i).$$

Assume for contradiction that

$$C < 1 + \frac{8(1+v)}{(1-v)^2}.$$

Let $K = C - 1$. Then

$$K(1-v)^2 < 8(1+v).$$

A direct calculation shows that the quadratic

$$Bs^2 + (1-A)s + 1$$

has negative discriminant, because

$$(1-A)^2 - 4B = \frac{K}{4} (K(1-v)^2 - 8(1+v)) < 0.$$

Hence

$$Bs^2 + (1-A)s + 1 > 0$$

for all s . Consequently,

$$f(s) > s$$

whenever $A - Bs > 0$. Since $r_i > 0$, the inequality $r_i \leq A - Bs_i$ forces $A - Bs_i > 0$, so $s_i < A/B$. Thus

$$s_{i+1} \geq f(s_i) > s_i.$$

So the sequence (s_i) is strictly increasing and bounded above by A/B . It must converge to some $L \leq A/B$. Passing to the limit in $s_{i+1} \geq f(s_i)$ gives

$$L \geq f(L).$$

But for every $L < A/B$, we have $f(L) > L$, and $L = A/B$ is not in the domain of f . This is a contradiction. Therefore

$$C \geq 1 + \frac{8(1+v)}{(1-v)^2}.$$

2.4.2 Discover Search.

2.4.3 Capture Search.

2.5 Oblivious Mobile Target: Knowledge Models

There are natural questions here concerning knowledge models.

3 Conclusion

References